Arrow's Impossibility Theorem

2026-01-14

Overview

Figure 1: Kenneth Arrow in 2010. Photograph shows him smiling, wearing glasses and a dark suit with patterned tie.

- Lived 1921-2017.
- Worked primarily at Stanford University.
- Won the Nobel Memorial Prize in Economic Sciences in 1972.
- Could have won it twice over: once for work on general equilibrium theory, and once for what we'll cover today.

Arrow's Impossibility Theorem (1951) demonstrates that no voting system can simultaneously satisfy a set of seemingly reasonable conditions for democratic decision-making.

Definition: An individual preference ranking is an ordering of all available alternatives from most preferred to least preferred, with ties allowed, but not gaps.

We'll write $A >_i B$ to mean that person i prefers A to B, and $A =_i B$ to mean that person i thinks that A and B are equally good.

Key Assumptions:

- **Completeness**: Exactly one of $A >_i B$, $B >_i A$, and $A =_i B$ holds.
- **Transitivity of Better**: If $A >_i B$ and $B >_i C$, then $A >_i C$.
- **Transitivity of Equal**: If $A =_i B$ and $B =_i C$, then $A =_i C$.

We'll write A $>_\forall$ B to mean that the group prefers A to B, and A $=_\forall$ B to mean that the group thinks they are equally good.

Same Key Assumptions

- **Completeness**: Exactly one of A $>_\forall$ B, B $>_\forall$ A, and A $=_\forall$ B holds.
- **Transitivity of Better**: If A $>_\forall$ B and B $>_\forall$ C, then A $>_\forall$ C.
- **Transitivity of Equal**: If A $=_\forall$ B and B $=_\forall$ C, then A $=_\forall$ C.

Big Question

How do we combine individual rankings into a single social ranking?

What We Want: A systematic method (social choice function) that takes individual preference profiles and produces a group preference ranking.

Background

Assumptions Again

There are a few unusual features of this setup:

- Most of us don't have views about all possible outcomes of an election.
- The combination just takes individual *rankings* into account. It doesn't look at *strength* of preference.
- The output doesn't just ask for a winner, but for a full ordering.

A Non-Answer

Say $A >_{\forall} B$ if more people have $A >_i B$ than have $B >_i A$, and $A =_{\forall} B$ if the numbers are the same.

This is a very simple kind of majoritarian rule.

You can probably see where this is going.

- Imagine that there are three options, A, B, and C.
- And there are three voters, X, Y, and Z.
- X has $A >_X B >_X C$.
- Y has $B >_Y C >_Y A$.
- Z has $C >_Z A >_Z B$.

Condorcet Again

- Since two people (X and Z) prefer A to B, we have A $>_\forall$ B.
- Since two people (X and Y) prefer B to C, we have B $>_\forall$ C.
- Since two people (Y and Z) prefer C to A, we have C $>_\forall$ A.
- And we've violated transitivity.

Plurality Again

Let's try a different solution.

- Ignore every part of $>_i$ except who is top.
- For each option, give it 1 point for every voter for which it's the top option.
- If the voter has two options tied for top, give them 1/2 point each, and so on for more ties.
- Rank the options by how many points they receive.

Plurality Again

- This is guaranteed to satisfy the assumptions.
- Any approach where each option gets a number, and you rank the numbers, will produce the right kind of ordering.
- But it has a very strange result.

Plurality Paradox

Imagine there are three options, A, B, and C, and 100 voters. On Monday the preferences are:

Table 1: Voter preferences on Monday

Voters	Preference
40	A > B > C
35	B > C > A
25	C > B > A

So A wins with 40 votes, and B is second with 35.

Plurality Paradox (continued)

On Tuesday, 10 people in the last group change their mind, just about B and C. So now we have:

Table 2: Voter preferences on Tuesday

Voters	Preference
40	A > B > C
45	B > C > A
15	C > B > A

Now B moves ahead of A in the social ranking, even though no one changed their mind about A and B. Is this odd?

The Theorem

Arrow's Approach

Rather than going through options one at a time, let's identify some principles.

- Start with a list of things we'd like the aggregation to be like.
- See what kind of things could meet all of the requirements on our list.

For a very natural set of requirements, there are **zero** aggregation rules that satisfy all of them.

- Any aggregation rule you pick will have some slightly odd feature.

Arrow's Requirements

Arrow identified four conditions that seem essential for any fair democratic system:

1. **Unrestricted Domain** (U)
2. **Weak Pareto Principle** (P)
3. **Independence of Irrelevant Alternatives** (I)
4. **Non-dictatorship** (D)

Unrestricted Domain (U)

What it means: The voting system must work for ANY possible combination of individual preference rankings.

Why it seems reasonable: We shouldn't restrict what preferences people are allowed to have in a democracy.

Weak Pareto Principle (P)

What it means: If everyone prefers alternative A to alternative B, then the group ranking should also prefer A to B.

Why it seems reasonable: If there's unanimous agreement, the social choice should reflect that agreement.

Figure 2: Vilfredo Pareto, circa 1870s.

- Lived 1848-1923.
- Trained in Turin, worked primarily at the University of Lausanne.
- He died around the time Mussolini came to power. The Italian fascists claimed him as an inspiration, though this *may* have been a misreading; he often sounds like a classical liberal.

Weak Pareto: If for all i, A $>_i$ B, then A $>_\forall$ B.

Strong Pareto: If for some i, A $>_i$ B, and for no i, B $>_i$ A (but maybe there are some ties), then A $>_\forall$ B.

The latter is more contentious, but it isn't needed for the proof.

(In fact, all that the proof really needs is that every outcome is possible.)

What it means: The social ranking between any two alternatives should depend *only* on how individuals rank those two alternatives, not on how they rank other alternatives.

Key Insight: If we're deciding between A and B, it shouldn't matter what people think about C.

Formal Definition: For any two preference profiles R and R', if every individual has the same preference between alternatives A and B in both profiles, then the social choice between A and B must be the same in both profiles.

This isn't a property of how the aggregation function treats what happens on any given day; it's a property of how it treats possible changes in individual preferences.

Non-dictatorship (D)

What it means: No single individual's preferences should always determine the group ranking, regardless of what others prefer.

Why it seems reasonable: This is basic to democratic ideals—no one person should have absolute power.

Non-dictatorship (D)

Formal statement: For every individual i, there is some profile R and some pair of options A and B such that in R, $A >_i B$, but not $A >_\forall B$.

Note: This doesn't prevent someone from being influential, just from being decisive in every case.

You might have wanted a stronger condition, like this one. But it turns out D is enough to get the result.

Anonymity (A) If two people switch their preferences, so on Tuesday X has the views that Y had on Monday, and Y has the views that X had on Monday, that shouldn't change the answer.

Arrow's Theorem: There is no social choice function that simultaneously satisfies:

- Unrestricted Domain (U);
- Weak Pareto Principle (P);
- Independence of Irrelevant Alternatives (I); and
- Non-dictatorship (D)

Proof

I'm not going to make you sit through a proof here. There are many proofs available if you want to find one (including in the SEP article).

The usual structure is to assume U, P, and I, and infer the existence of a dictator.

The main benefit of sitting through a proof is that you see just how strong a condition I is, and we'll come back to that strength next time.

For Next Time

We'll look at ways out of, or around, the problem. Some options include:

- Restricting the domain (dropping U);
- Using more information than just ordinal rankings;
- Accepting certain violations of IIA;
- Single-peaked preferences and median voter theorems.